Exercise 03: Individual Proposal for Course Project

Multimodal AI for Muscular Failure Prediction in Physical Training

- **Course:** Computational Thinking (CS252 - Class 25A01)
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- **Proposed Project Domain:** Biomechanics, Multimodal Machine Learning & Computational Ergonomics
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1. Executive Summary

In high-intensity resistance training, pushing sets close to momentary muscular failure is essential for maximizing neuromuscular adaptation and hypertrophic stimulus. However, accurately judging proximate failure—formulated in exercise science as **Reps in Reserve (RIR)** or **Rating of Perceived Exertion (RPE)**—is notoriously difficult for both novices and experienced lifters. Misjudging exhaustion frequently leads to catastrophic mechanical failure, uncontrolled drops under heavy free weights (e.g., bench press, squats, dips), and severe musculoskeletal injury.

This proposal introduces a **Multimodal AI System for Real-Time Muscular Failure Prediction**. By synthesizing high-frequency computer vision kinematics (barbell velocity loss, joint angular deceleration, and form breakdown) with wearable sensor telemetry (smartwatch inertial micro-tremors and surface electromyography signals), the system computes real-time physiological exertion curves, dynamically predicts remaining Reps in Reserve ($RIR \in [0, 3]$), and issues anticipatory safety alerts before catastrophic failure occurs.

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2. Problem Statement & Motivation

The Limitation of Unimodal Vision Systems

Existing commercial AI fitness applications (such as Apple Fitness+, Peloton Guide, or standard pose-estimation mobile apps) rely exclusively on single-camera computer vision to extract 2D/3D joint coordinates. While spatial keypoint models (e.g., MediaPipe Pose, YOLO-Pose) are effective for verifying geometric form (such as checking whether knees cave during squats or the lower back arches during deadlifts), **vision alone cannot measure internal muscular exertion, motor unit recruitment fatigue, or central nervous system exhaustion**.

In compound resistance exercises such as heavy barbell rows, dips, squats, and overhead presses:

1. **Geometric Invariance Prior to Failure:** A disciplined lifter often maintains flawless geometric posture right up to the penultimate repetition. The subsequent repetition suffers sudden mechanical failure mid-concentric phase, leaving standard vision models zero lead time to warn the user.
2. **Subjective RIR Inaccuracy:** Research in sports science demonstrates that lifters misjudge their actual RIR by an average of 1.5 to 2.5 repetitions, resulting in either sub-optimal training stimulus or dangerous overtraining.

3. **Absence of Intelligent Spotting:** Solo lifters without human spotters face real physical hazards when training to failure in home gyms or crowded fitness centers.

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3. Independent Value & Differentiation from Group Project

In compliance with the assignment requirement for independent thinking:

- **Group Project Scope:** The collaborative group project focuses primarily on broad health informatics, basic workout session logging, and routine posture classification using standard web dashboards.
- **Individual Proposal Differentiation:**
  - **Distinct Problem Domain:** Real-time neuromuscular fatigue dynamics and biomechanical failure boundary prediction rather than static form evaluation.
  - **Novel Technical Approach:** Multimodal sensor fusion coupling high-speed optical kinematics ($60\ \text{fps}$ video) with inertial measurement units (IMU high-frequency accelerometer/gyroscope signals) to capture sub-visual micro-tremors ($8 - 12\ \text{Hz}$).
  - **Safety-Critical Feature Set:** Dynamic RIR regression and an anticipatory audio-haptic emergency warning trigger when the system detects imminent concentric stall, effectively functioning as an autonomous digital spotter.

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4. Conclusion

The Multimodal Muscular Failure Prediction system bridges the critical gap between visual movement aesthetics and internal muscular physiology. By grounding safety-critical training decisions in computational thinking, multimodal sensor fusion, and predictive machine learning, the project delivers a novel, practical, and intellectually rigorous contribution to sports engineering and injury prevention.